

Technical Document #338: Natural Language Processing - Comprehensive Overview

Technical Document #338: Natural Language Processing - Comprehensive Overview

Coreference resolution identifies all expressions that refer to the same entity in a text. It is essential for understanding document-level coherence.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Coreference resolution identifies all expressions that refer to the same entity in a text.
- Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names.
- Tokenization is the process of breaking text into individual units called tokens.